

Notes: Staiger and Tabellini (AER, 1987)
Discretionary Trade Policy and Excessive Protection

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1 Big picture

1.1 Question

Why do small countries often use tariffs and other trade-distorting policies, even though classical trade theory says that free trade is optimal for a small country and that, if redistribution is desired, production or consumption subsidies should dominate tariffs?

Staiger and Tabellini propose an answer based on **time inconsistency**. A government with redistributive motives may want to protect an import-competing sector after an adverse terms-of-trade shock. The problem is that the redistributive power of protection depends on whether workers anticipated it before moving across sectors.

1.2 Core answer

The paper's core idea is simple:

- A tariff announced in advance affects workers' relocation decisions.
- If workers expect protection, fewer workers leave the injured import sector.
- Because labor does not fully move away, the announced tariff becomes less useful as a redistributive surprise.
- After labor allocation is fixed, the government has an incentive to impose more protection than workers expected.
- Rational workers anticipate this incentive, so the equilibrium tariff is positive even when free trade is the optimal commitment policy.

Interpretation: The government would like to commit to free trade, but once labor has moved and the shock is realized, it has a temptation to use tariffs to redistribute income. Since private agents understand this temptation, free trade loses credibility. The resulting discretionary policy is overprotective.

1.3 Main takeaways

- The paper gives a **positive theory of excessive protection** without relying on lobbying or political capture.
- Free trade can be optimal under commitment but not time consistent.
- The time-consistent equilibrium involves protection that is socially costly and does not improve the income distribution relative to free trade.
- A tariff can welfare-dominate a time-consistent production subsidy, reversing the traditional static ranking of trade policy instruments.
- International trade agreements, such as GATT-style commitments, can be interpreted as devices that help governments commit against their own future discretionary temptations.

2 Incremental contribution of the paper

2.1 What the paper adds relative to prior trade theory

Traditional small-country trade theory argues that free trade is optimal. If a government nevertheless wants to redistribute income toward an injured sector, the standard policy-ranking result says that targeted subsidies should dominate tariffs because tariffs distort both production and consumption.

This paper adds a dynamic-incentive qualification: the relevant comparison is not only between static instruments, but between **credible policies**. If the government cannot commit, the policy that would be optimal under commitment may not be implementable.

2.2 What the paper adds relative to political economy models

Earlier political economy models explain protection using lobbying, special interests, or redistributive political weights. Staiger and Tabellini deliberately abstract from those forces. Their government is benevolent and has social welfare objectives, but protection still emerges.

Incremental point: Excessive protection can arise from the interaction between discretionary policy and private-sector expectations, even without lobbying.

2.3 What the paper adds relative to time-consistency theory

The paper imports the Kydland-Prescott logic of rules versus discretion into trade policy. The government would like the private sector to believe one policy before labor reallocates, but after labor has reallocated the government wants to do something else.

New mechanism: In this model, the incentive to deviate comes from the greater redistributive impact of **unexpected** protection. Surprise tariffs alter relative wages after workers have already made costly relocation choices.

2.4 What the paper adds for policy design

The model gives a rationale for binding trade agreements that is different from standard international cooperation. GATT/WTO-type commitments are not only ways to constrain beggar-thy-neighbor policies across countries. They can also help a country constrain its own government in its strategic game with domestic residents.

Practical implication: A simple rule like free trade can dominate discretionary activism because discretion undermines credibility.

3 Model environment

3.1 Basic economy

The model considers a small open economy with two traded goods and two inputs:

- good x : export good;
- good y : import good;
- capital: immobile across sectors;
- labor: mobile across sectors, but mobility is costly.

Both sectors have the same production function:

$$i = f(N^i), \quad i = x, y, \quad f'(\cdot) > 0, \quad f''(\cdot) < 0, \quad (1)$$

where N^i is labor employed in sector i .

The economy faces an adverse terms-of-trade shock: the world price of imports y falls by ε . The government can impose an ad valorem tariff t on imports.

3.2 Wage conditions

Firms hire labor until the value of marginal product equals the wage:

$$p^x f'(N^x) = W^x, \quad (2)$$

$$p^y(1+t)f'(N^y) = W^y. \quad (3)$$

Before the shock, labor is equally allocated across sectors and free trade prevails. After the shock, workers in the injured import sector y may move to the export sector x .

3.3 Costly labor mobility

A worker who moves from sector y to sector x loses productivity. Let $\lambda \in (0, 1)$ measure the fraction of productivity retained by movers. If λ is lower, reallocation is more costly.

Let α be the fraction of labor that remains in sector y after reallocation. Then effective labor in each sector is:

$$N^y = \alpha, \quad N^x = (1 - \alpha)\lambda + 1. \quad (4)$$

The expected wage ratio must satisfy:

$$\frac{1}{\lambda} \geq \frac{W^{y,e}}{W^{x,e}} \geq \lambda. \quad (5)$$

Interpretation: Workers move only if the expected wage gain is large enough to compensate for lost productivity in the destination sector.

3.4 Timing and the credibility problem

The crucial timing assumption is that, absent a binding commitment, the government cannot irreversibly set tariff policy before labor reallocates. Workers choose whether to move based on the **expected** tariff t^e , while the government later chooses the **actual** tariff t .

This creates the time-consistency problem:

- If the government could commit, workers would believe the announced tariff.
- If the government cannot commit, workers know that after they relocate the government may want to change the tariff.
- The actual policy must therefore be a rational-expectations, time-consistent policy.

4 Reallocation of labor and the role of expected protection

4.1 Expected tariff and labor allocation

The expected wage ratio after the shock is:

$$\frac{W^{y,e}}{W^{x,e}} = \frac{(1 + t^e)(1 - \varepsilon)f'(\alpha)}{f'((1 - \alpha)\lambda + 1)}. \quad (6)$$

Expected protection affects labor reallocation. A higher expected tariff raises the expected return to staying in the injured import sector, so fewer workers leave sector y .

The key lemma says:

- For moderate expected tariffs, workers relocate until the wage ratio equals λ .
- In that range, expected protection changes α but does not change the equilibrium relative wage.
- If the expected tariff is sufficiently high, no workers move; then relative wages can rise above λ .

Interpretation: Anticipated tariffs are partly undone by labor allocation. This is why the government may prefer surprise protection: surprise tariffs change wages after the labor allocation is already fixed.

5 Optimal tariff policy under commitment

5.1 Government objective

The government maximizes a social welfare function over workers' indirect utilities. It cares about redistribution because marginal utility of income is diminishing. After a negative shock to the import sector, workers remaining in sector y have high marginal utility of income, so the government values transfers toward them.

However, tariffs create two costs:

- consumption distortions, because tariffs drive a wedge between domestic and world prices;
- production distortions, because anticipated protection affects labor allocation.

5.2 Optimal policy

With commitment, the government chooses a tariff that workers fully anticipate. The paper shows that the optimal tariff policy is either:

- free trade, $t = 0$; or
- a sufficiently high tariff that prevents any labor reallocation.

For the central case studied in the paper - when relocation costs are not too large - **free trade is optimal under commitment**.

Interpretation: When tariffs are anticipated, small positive tariffs do not generate marginal redistributive gains in the moving range because workers adjust their location. But they still create distortions. Therefore, the best commitment policy is free trade.

6 Time-consistent policy and excessive protection

6.1 Why free trade is not credible

Suppose the government announces free trade. Workers believe $t^e = 0$ and relocate accordingly. After this relocation has occurred, the government takes α as fixed. A surprise tariff now raises the income of workers remaining in the injured import sector and redistributes toward workers with high marginal utility of income.

At $t = t^e = 0$, the marginal benefit of a surprise tariff is strictly positive, while the marginal cost is initially zero. Therefore, the government has an incentive to deviate from free trade.

6.2 Time-consistent equilibrium

The time-consistent equilibrium is defined by two conditions:

1. the tariff is fully expected, so $t = t^e$;
2. given workers' expectations and the resulting labor allocation, the government has no incentive to choose a different tariff.

The resulting tariff is positive:

$$0 < \hat{t} < \frac{\lambda}{1 - \varepsilon} - 1. \quad (7)$$

This is the main result:

When free trade is optimal under commitment, it is not time consistent. The time-consistent policy involves a socially excessive level of protection.

Interpretation: The discretionary tariff does not improve the final wage distribution relative to free trade in the moving range, because workers anticipate it and adjust. The country gets the trade distortion without the redistributive gain that motivated the policy in the first place.

7 Tariffs versus production subsidies

7.1 Traditional ranking

In standard welfare analysis, if the government wants to help the injured sector, a production subsidy should dominate a tariff because it targets production without distorting consumption.

7.2 Time-consistent ranking

Staiger and Tabellini show that this ranking can reverse once credibility is imposed.

A time-consistent production subsidy is set at a high level, high enough to equalize wages across sectors:

$$1 + \hat{s} = \frac{1}{1 - \varepsilon}. \quad (8)$$

The time-consistent tariff is lower than the time-consistent subsidy:

$$1 + \hat{t} < \frac{1}{1 - \varepsilon} = 1 + \hat{s}. \quad (9)$$

The subsidy avoids consumption distortions, but because it is higher, it can generate larger production distortions. If the social benefits of redistribution are small enough, the tariff can welfare-dominate the production subsidy.

Interpretation: Static efficiency says subsidies dominate tariffs. Dynamic credibility says a government may prefer the tariff because the credible subsidy must be too large.

8 Tables and figures

8.1 Figures 1 and 2: anticipated versus surprise protection

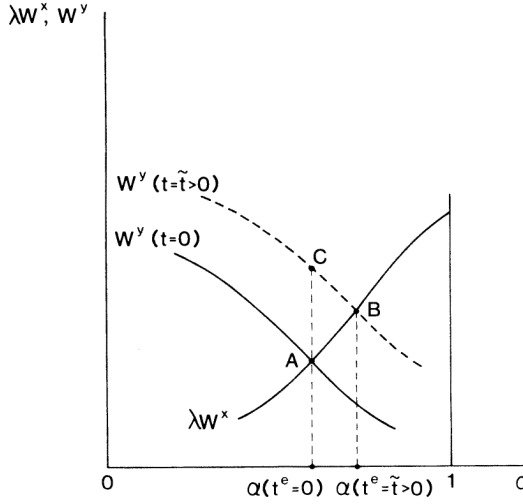


FIGURE 1

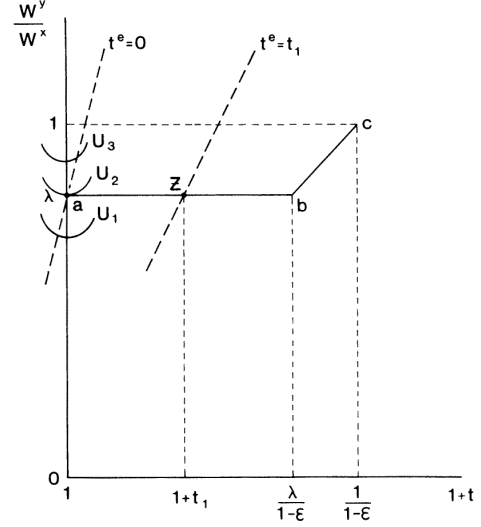


FIGURE 2

(a) Figure 1. Labor reallocation and surprise protection.

(b) Figure 2. Equilibrium locus and off-equilibrium surprise paths.

Original figures extracted from Staiger and Tabellini (1987).

Read:

- **Figure 1** plots wages in the import sector and export sector as labor moves. With no expected tariff, equilibrium is at point *A*: workers leave sector *y* until the wage gain equals the mobility cost.
- An anticipated tariff shifts the wage schedule for sector *y* and moves the equilibrium to point *B*. Fewer workers leave the injured sector, but the wage differential is still offset by labor reallocation.
- An unexpected tariff moves the economy toward point *C*: labor allocation is unchanged, so the tariff directly redistributes income toward workers remaining in sector *y*.
- **Figure 2** maps the feasible rational-expectations combinations of tariffs and relative wages. The locus *abc* gives fully anticipated policy outcomes. Dashed lines show outcomes if the actual tariff differs from expected tariffs.

Economic message: The difference between anticipated and unexpected protection is the core of the paper. Anticipated tariffs change labor allocation and lose redistributive bite; unexpected tariffs redistribute more powerfully. This creates the government's temptation to surprise and destroys the credibility of free trade.

8.2 Table 1 and Figure 3: time-consistent protection and welfare comparisons

TABLE 1 — TARIFFS AND SUBSIDIES COMPARED

Computations ^a							
	$\hat{t}$	$\alpha(\hat{t})$	$J(\hat{t})$	$\hat{s}$	$J(\hat{s})$	$\alpha(\bar{t}=0)$	$J(\bar{t}=0)$
$\lambda = .75$	.3	.92	1.7561	1	1.7835	.78	1.7820
$\lambda = .9$	.08	.74	1.815	1	1.7835	.70	1.818

^aThe computations are based on the following assumptions. The utility function is $U = \frac{1}{2} \ln x + \frac{1}{2} \ln y$. The production function is $x = \ln N^x$; $y = \ln N^y$. The labor force in each sector prior to the realization of the shock has been normalized to equal 10. Finally, the size of the shock to the world price of y has been taken to be $\epsilon = .5$. The symbols in the table are defined as follows: $\bar{t} \equiv$ optimal tariff; $\hat{t} \equiv$ time-consistent tariff; $\alpha \equiv$ fraction of the labor force remaining in sector y , as a function of the tariff; $\hat{s} \equiv$ time-consistent subsidy; $J(\cdot) \equiv$ value taken in equilibrium by the social welfare function, as a function of the various policy instruments; $\lambda \equiv$ ratio between the marginal product of the labor that has just moved to sector x from sector y , and the marginal product of the labor originally in sector x that remains there.

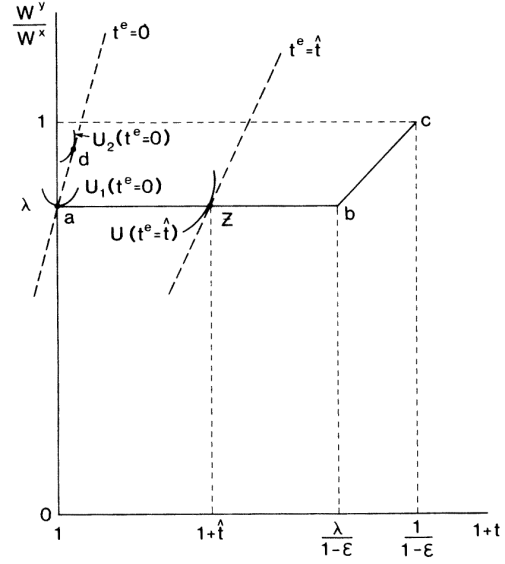


FIGURE 3

(a) Table 1. Tariffs and subsidies compared.

(b) Figure 3. Time-consistent tariff equilibrium.

Original table and figure extracted from Staiger and Tabellini (1987).

Read:

- **Table 1** provides numerical examples using logarithmic utility and production. When $\lambda = .75$, the time-consistent tariff is 30%, the time-consistent subsidy is 100%, and the subsidy gives higher welfare than the tariff. When $\lambda = .9$, the time-consistent tariff falls to 8%, while the time-consistent subsidy remains 100%; in this case, the tariff welfare-dominates the subsidy.
- In both examples, the free-trade commitment benchmark gives higher welfare than the time-consistent tariff.
- **Figure 3** shows the government's temptation at free trade. Starting from point a with $t^e = 0$, the government would like to move along the dashed line to a surprise-protection point d .
- Because workers anticipate this incentive, the actual equilibrium is point z on the rational-expectations locus, where the positive tariff $\hat{t}$ is fully expected.

Economic message: Table 1 shows that the credibility constraint can reverse the static ranking of policy instruments. Figure 3 shows why the positive tariff is an equilibrium trap: once workers expect discretionary protection, the government validates that expectation, leaving the economy with inefficient trade distortion.

9 Mechanism in one diagrammatic logic

The model can be summarized as a four-step chain:

1. A negative terms-of-trade shock hurts the import sector.
2. Workers decide whether to leave the import sector based on expected protection.
3. If the government cannot commit, it later wants to use surprise protection to redistribute toward injured workers.
4. Rational workers anticipate this, so the equilibrium tariff is positive and expected, even though the commitment optimum may be free trade.

Key distinction: Commitment policy asks, “What would the government want workers to expect?” Discretionary policy asks, “What will the government actually do after workers have moved?” The answers differ.

10 Why the result matters

10.1 Positive explanation of protection

The paper explains protection as a credibility problem, not merely a political-interest problem. A benevolent government can be trapped into protection because it cannot credibly promise not to redistribute ex post.

10.2 Rules versus discretion in trade policy

The model supports simple rules even when the government is benevolent. Discretion can be costly because it changes private expectations and induces inefficient equilibrium behavior.

10.3 Institutional role of trade agreements

The analysis suggests an important domestic role for international commitments. A trade agreement can help governments bind themselves to future free trade, thereby making policy announcements credible to domestic workers and firms.

10.4 Instrument choice under credibility constraints

The paper also helps explain why governments might use tariffs instead of production subsidies. When the government must choose a time-consistent policy, the standard static efficiency ranking may no longer apply.

11 Limitations and interpretation

- The model is stylized: two sectors, one shock, fixed capital, and a simplified form of costly labor mobility.
- The government is benevolent, so the model omits lobbying and interest-group politics. This is a strength for isolating the time-consistency mechanism, but it is not a full model of real-world trade politics.
- The analysis is mostly theoretical and illustrative; Table 1 provides numerical examples, not empirical calibration.
- Labor reallocation is modeled in a one-period way. The authors note that an intertemporal extension with learning-by-doing or reputation would modify, but not eliminate, the core logic.

Interpretation: The paper should be read as a mechanism paper. It isolates one reason why protection can be excessive: discretionary policy interacts with private expectations. The model is not meant to replace political economy explanations, but to add a credibility channel to them.

12 Short summary

Staiger and Tabellini show that free trade can be optimal but not credible. After an adverse terms-of-trade shock, a government with redistributive objectives wants to use protection to help workers in the injured import sector. If the tariff is anticipated, workers adjust their location and the redistributive effect is weakened. If the tariff is unexpected, it is more redistributive. Therefore, once labor allocation is fixed, the government wants to surprise the private sector with protection. Rational workers anticipate this, so the time-consistent equilibrium has a positive tariff. The tariff is excessive because it creates trade distortions without delivering the intended redistributive advantage relative to free trade. The model also shows that, under time consistency, tariffs may dominate production subsidies, reversing the standard static ranking.

13 Conclusion

The paper's central contribution is to connect trade protection to the rules-versus-discretion problem. Its main message is that credible commitment matters for trade policy. A government may know that free trade is best, but if it cannot bind its future self, private agents will expect protection and the government will validate that expectation. This produces excessive protection and helps explain why binding trade agreements can improve welfare by limiting domestic discretion.